

INTERNET OF THINGS
MODULE – 2
IoT Sensing and Actuation & Domain Specific IoTs
QUESTION BANK

S. No	Question	CO	BTL	Marks
1	Discuss the characteristics of actuators and compare electrical, hydraulic, and pneumatic actuators.	CO2	L2 (Understand) + L4 (Analyze)	10M
2	Explain the importance of sensors and actuators in IoT, citing real-world applications.	CO2	L2 (Understand)	10M
3	Discuss various sensing types and their roles in IoT applications across multiple domains.	CO2	L2 (Understand)	10M
4	Explain the challenges in sensor calibration, selection, and maintenance in IoT systems.	CO2	L2 (Understand) + L4 (Analyze)	10M
5	Explain IoT applications in the healthcare sector, including wearables and remote patient monitoring.	CO2	L2 (Understand)	10M
6	Compare IoT applications in agriculture and industry with suitable examples.	CO2	L2 (Understand) + L4 (Analyze)	10M
7	Discuss IoT applications in smart energy systems with examples of smart grids and smart meters.	CO2	L2 (Understand)	10M
8	Discuss domain-specific IoT applications in any four areas (Home Automation, Retail, Logistics, Health, Agriculture).	CO2	L2 (Understand)	10M
9	With suitable case studies, explain IoT applications in smart cities.	CO2	L3 (Apply)	10M
10	Describe IoT applications in the retail sector with real-world use cases.	CO2	L2 (Understand)	10M
11	With a neat diagram, explain how sensors and actuators work together in an IoT system.	CO2	L2 (Understand) + L3 (Apply)	8M
12	Describe different types of actuators with suitable real-world applications.	CO2	L2 (Understand)	8M
13	Discuss different types of sensing methods used in IoT (physical, chemical, biological).	CO2	L2 (Understand)	8M
14	Define a sensor and explain any two of its characteristics with examples.	CO2	L1 (Remember) + L2 (Understand)	5M
15	What is hysteresis in sensors? Illustrate with a diagram.	CO2	L1 (Remember) + L2 (Understand)	5M
16	Differentiate between sensors and actuators with examples.	CO2	L4 (Analyze)	5M
17	List and explain any three types of sensor deviations.	CO2	L1 (Remember) + L2 (Understand)	5M
18	Write short notes on pneumatic actuators with applications.	CO2	L2 (Understand)	5M
19	List any two IoT applications in home automation.	CO2	L1 (Remember)	5M
20	Define smart grids and explain how they help in energy management.	CO2	L1 (Remember) + L2 (Understand)	5M
21	Explain shipment monitoring in logistics using IoT.	CO2	L2 (Understand) + L3 (Apply)	5M
22	Mention two IoT applications in healthcare and lifestyle.	CO2	L1 (Remember)	5M